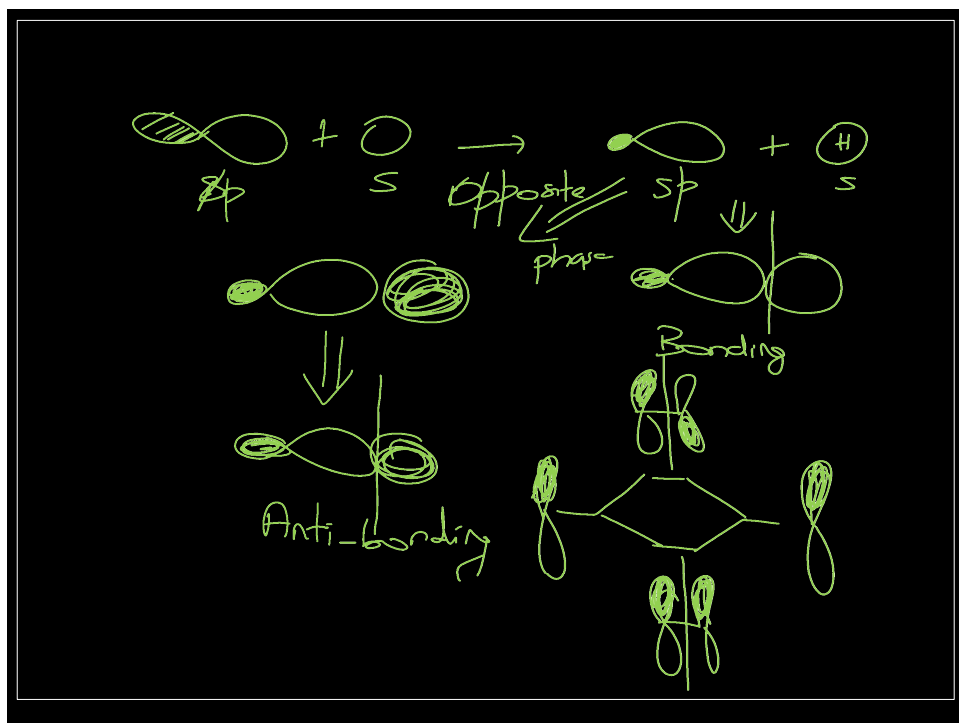
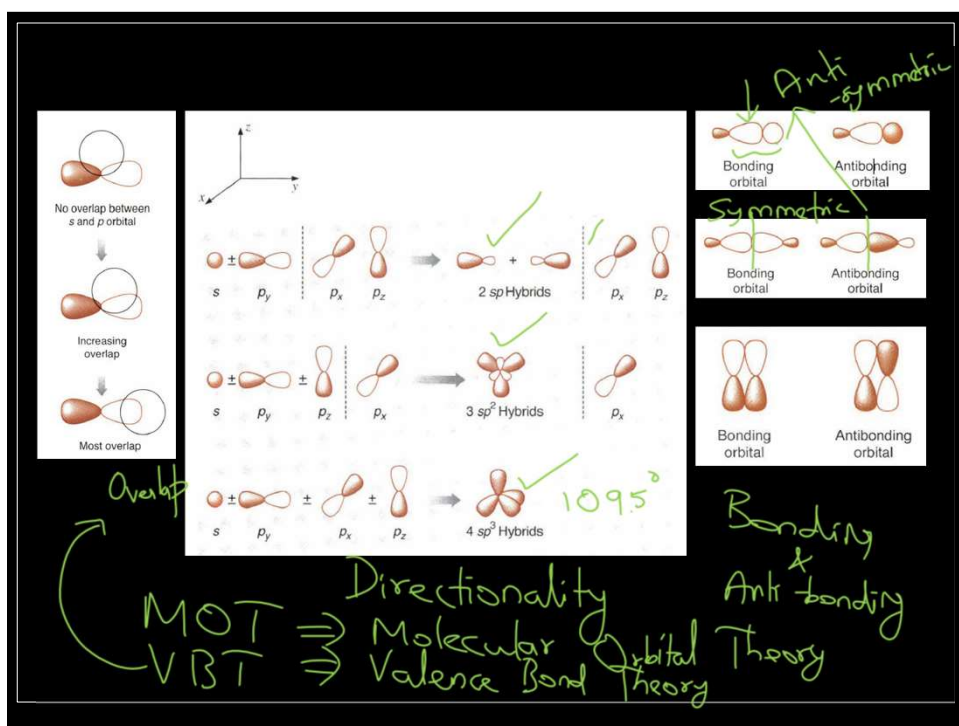
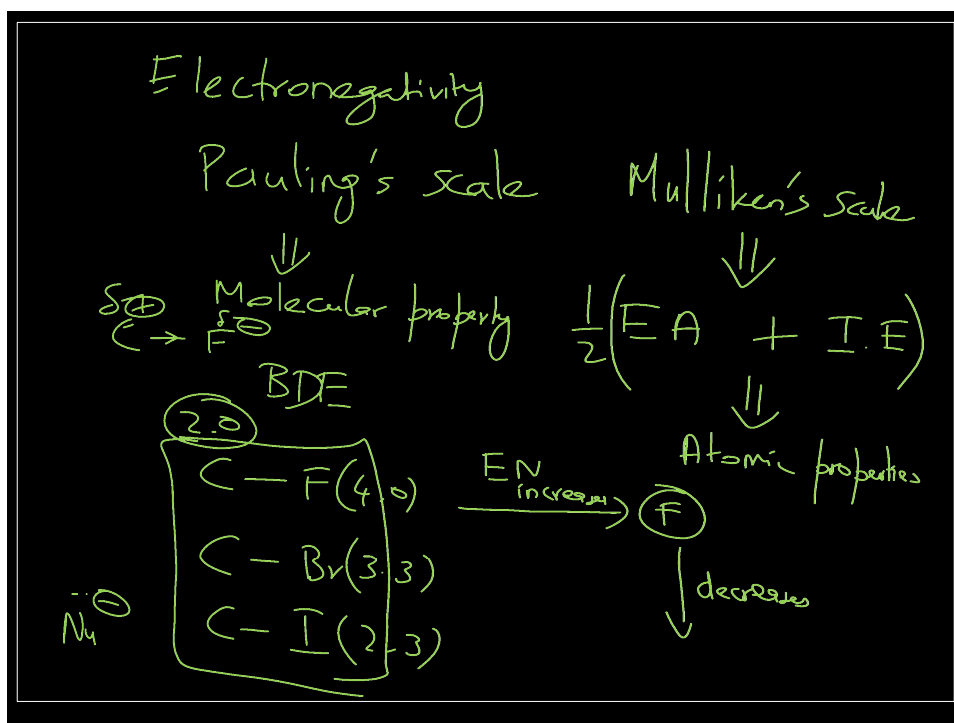
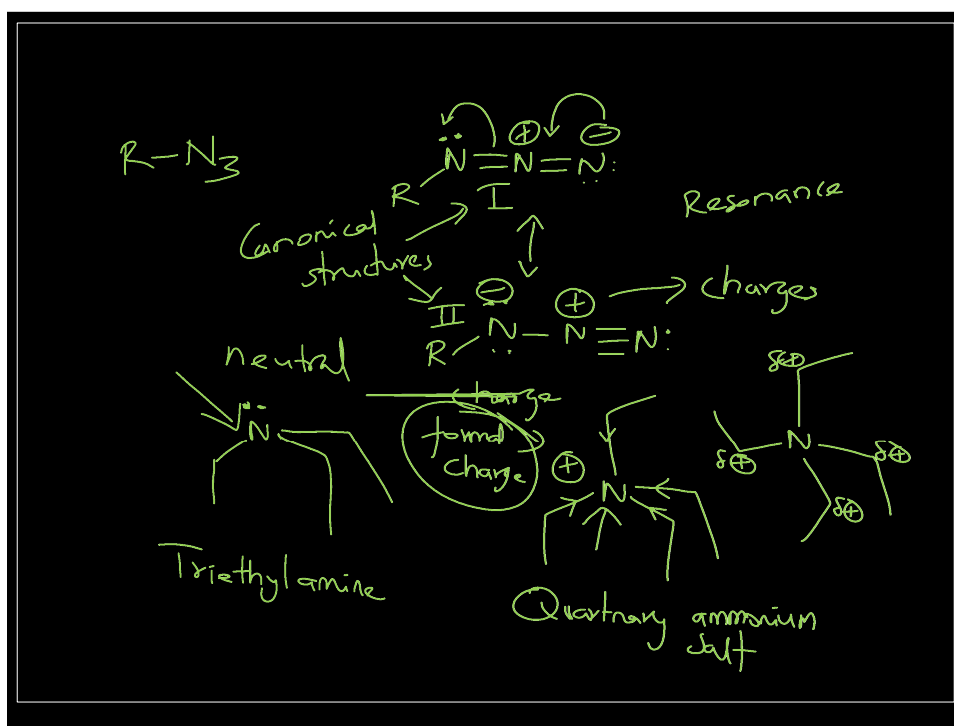






Concepts involving acidity, basicity, tautomerism, inductive, mesomeric, Nathan field and steric effects

Electronegativity:

- Useful to predict the charge distribution in an organic molecule.
- Linus Pauling original concept: "the power of an atom in a molecule to attract electrons to itself"
- Pauling assigned values to various atom types based on bond dissociation energies of molecules.
- Pauling electronegativity scale depends upon *molecular* properties
- Mulliken's electronegativity scale: "derived from the average of the ionization potential (the energy required to remove an electron from an atom or molecule) and electron affinity (the amount of energy released or required to attach another electron to an atom or molecule) of an atom"
- Mulliken's electronegativity is an *atomic* property.

Table 1.1
Electronegativities of Atoms According to the Scales of Pauling and Mulliken*

Atom	Pauling	Mulliken
H	2.1	3.01
B	2.0	1.83
C	2.5 ✓	2.67
N	3.0	3.08
O	3.5	3.22
F	4.0 ✓	4.44
Cl	3.0	3.54
Br	2.8	3.24
I	2.5	2.88
Li	1.0	1.28
Na	0.9	1.21
K	0.8	1.03
Mg	1.2	1.63
Ca	1.0	1.30
Al	1.5	1.37
Si	1.8	2.03
P	2.1	2.39
S	2.5	2.65

*Pauling, L. (1960). *The Nature of the Chemical Bond and the Structure of Molecules and Crystals; an Introduction to Modern Structural Chemistry*, 3d ed., Cornell University Press, Ithaca, NY, and Allen, L. C. "Electronegativity is the Average One-Electron Energy of the Valence-Shell Electrons in Ground-State Free Atoms." *J. Am. Chem. Soc.*, **111**, 9003 (1989).

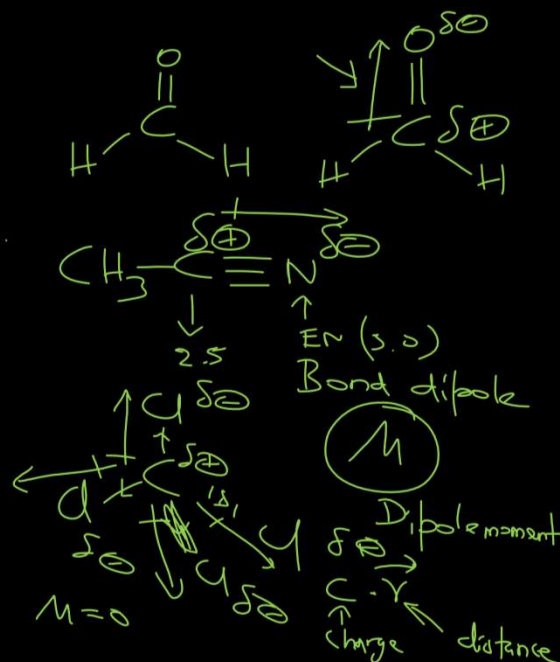
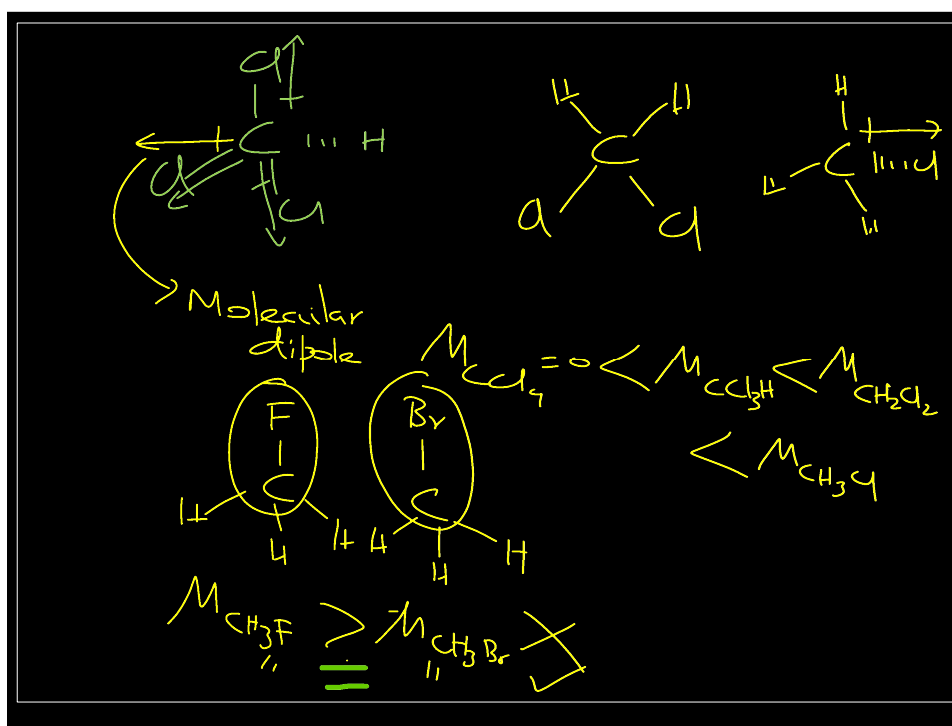


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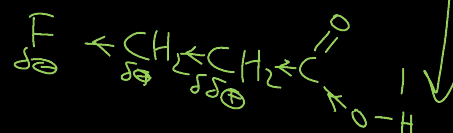


Handwritten notes on molecular dipole moments:

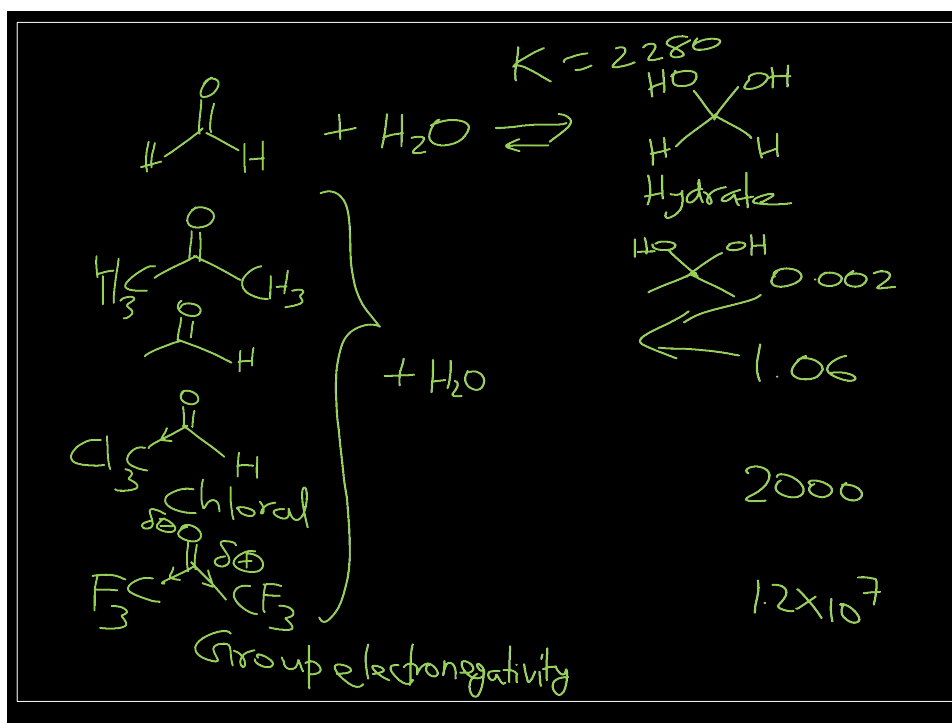
- Diagram of CH_3CN showing bond dipoles and the resultant molecular dipole.
- Diagram of CH_3F showing bond dipoles and the resultant molecular dipole.
- Note: δ^+ and δ^- charges on the atoms.
- Inequality: $M_{\text{CH}_3\text{F}} > M_{\text{CH}_3\text{Br}}$
- Inequality: $M_{\text{CCl}_4} = 0 < M_{\text{CCl}_3\text{H}} < M_{\text{CH}_2\text{Cl}_2} < M_{\text{CH}_3\text{Cl}}$

Compound	Molecular dipole (D)	Compound	Molecular dipole (D)
CCl_4	0.0	CH_3COCH_3	2.9
CHCl_3	1.0	CH_3COOH	1.7
CH_2Cl_2	1.6	CH_3COCl	2.7
CH_3Cl	1.9	$\text{CH}_3\text{COOCH}_3$	1.7
CH_3F	1.8	$\text{C}_6\text{H}_5\text{Cl}$	1.8
CH_3Br	1.8	$\text{C}_6\text{H}_5\text{NO}_2$	4.0
CH_3I	1.6	1-Butene	0.34
CH_3OH	1.7	1-Propyne	0.80
CH_3OCH_3	1.3	<i>cis</i> -2-Butene	0.25
CH_3CN	4.0	<i>cis</i> -1,2-Dichloroethene	1.9
CH_3NO_2	3.4	Tetrahydrofuran	1.6
CH_3NH_2	1.3	Water	1.8

Inductive effect
 $\text{CH}_3\text{CH}_2\text{COOH}$



acidity
increases



Inductive effect

An experimentally observable effect (on rates of reaction, etc.) of the transmission of charge through a chain of atoms by electrostatic induction. (IUPAC Goldbook)

ENG
-I

EDG
+I

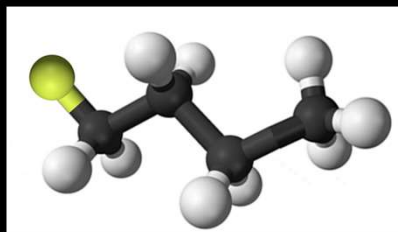
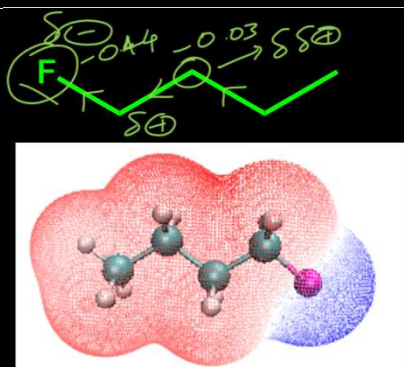
Group Electronegativities, Scaled to be Compatible with the Pauling Scale*

Group	Electronegativity
CH ₃	2.3
CH ₂ Cl	2.8
CHCl ₂	3.0
CCl ₃	3.0
CF ₃	3.4
Ph	3.0
CH=CH ₂	3.0
C=CH	3.3
C≡N	3.3
NH ₂	3.4
NH ₃ ⁺	3.8
NO ₂	3.4
OH	3.7

TABLE 1.3. Field Effects of Various Groups Relative to Hydrogen^a

+I		-I	
O ⁻	NR ₃ ⁺	COOH	OR
COO ⁻	SR ₂ ⁺	F	COR
CR ₃	NH ₃ ⁺	Cl	SH
CHR ₂	NO ₂	Br	SR
CH ₂ R	SO ₂ R	I	OH
CH ₃	CN	OAr	C≡CR
D	SO ₂ Ar	COOR	Ar
			C≡CR ₂

^aThe groups are listed approximately in order of decreasing strength for both -I and +I groups.



Blue ⊖
Red ⊕

